CS & IT ENGINEERING



COMPUTER ORGANIZATION AND ARCHITECTURE

CPU & Control Unit

Lecture No.- 2



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Recap of Previous Lecture



Topic CPU

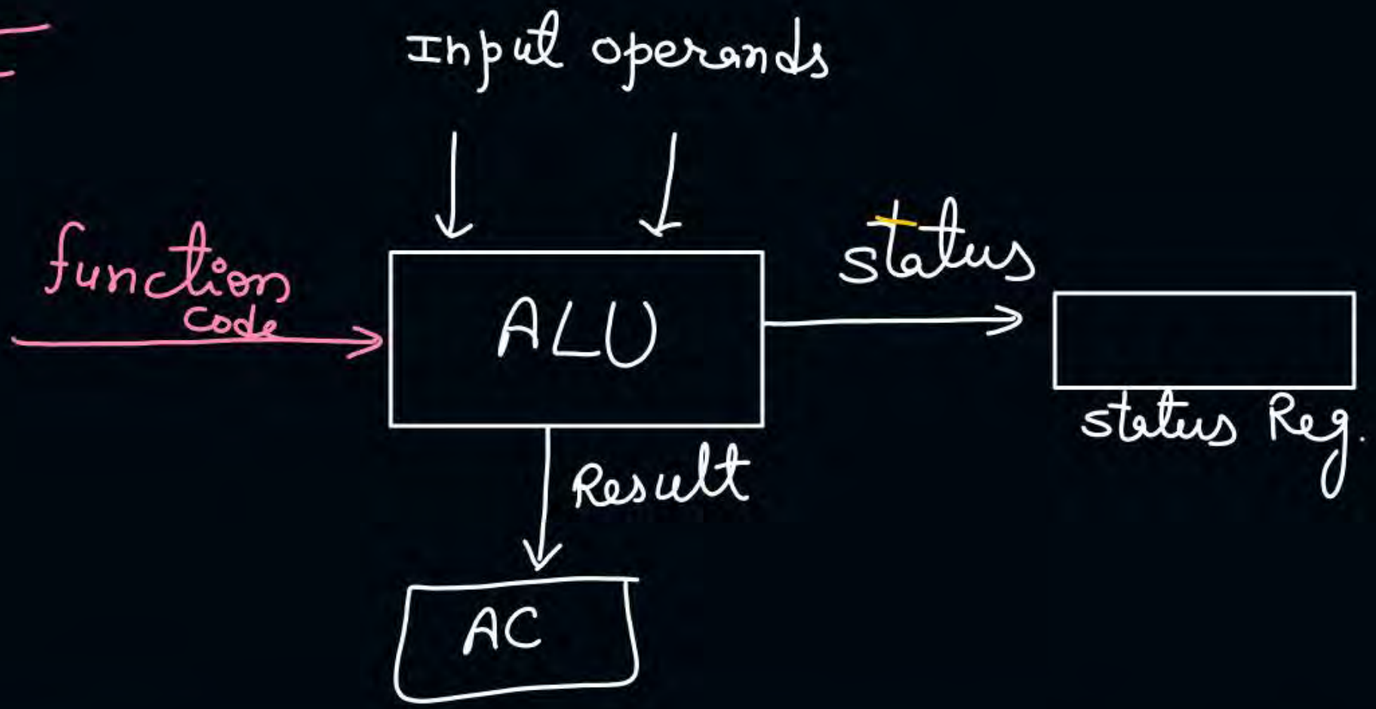
Topic MIPS

Topics to be Covered



- **Topic** Datapath
- **Topic** Control Unit Organization
- **Topic** Hardwired Control Unit

ALU :-



If ALU can perform ⇒ 16 operations

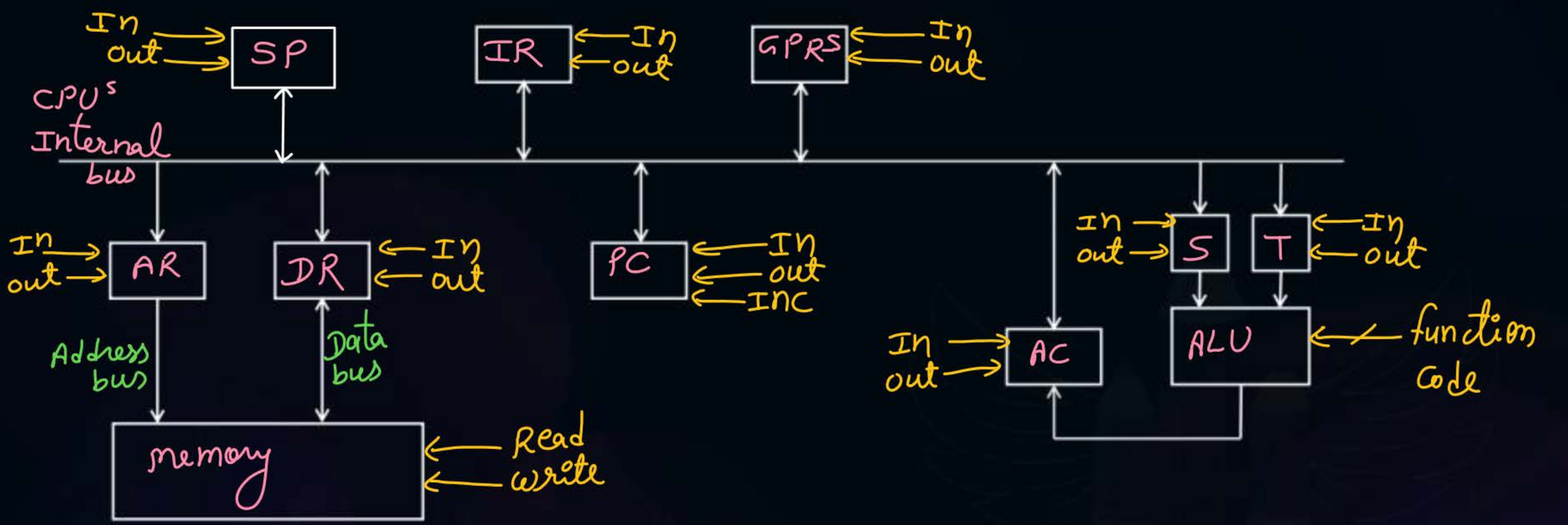
function code ⇒ 4 bits

Datapath :-

Collection of registers, ALU and other components to perform operations in CPU.



Topic : Datapath



Instruction fetch:-

1. $AR \leftarrow PC$ — PC_{out}, AR_{in}
2. $DR \leftarrow M[AR]$ — AR_{out}, $MEMORY_{Read}$, DR_{in}
3. $IR \leftarrow DR$, $PC \leftarrow PC + 1$ — DR_{out}, IR_{in}, PC_{inc}

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operand fetch to write back of $R1 \leftarrow R2 + R3$

operand fetch:
1. $S \leftarrow R2$ — $R2_{out}$, S_{in}
2. $T \leftarrow R3$ — $R3_{out}$, T_{in}

execution:
3. $AC \leftarrow S + T$ — S_{out}, T_{out}, ALU_{ADD}, AC_{in}

write back:
4. $R1 \leftarrow AC$ — AC_{out}, $R1_{in}$

Questⁿ Type 1 :-

No. of cycles needed for inst^n fetch, if mem. access operations take 4 cycles, and remaining all take 1 cycle.

1. AR ← PC ——— 1
2. DR ← M[AR] ——— 4
3. IR ← DR, PC ← PC+1 ——— 1

} Total = 6

Ques 2) No. of cycles for R1 ← R2 + R3

Ans = 4 (one for each microoperation)



Topic : Control Unit

It generates control signal and sends them to all components.
Components perform operations according to those signal.

Control variable :-
Name or label of a control signal

Control word :-
Collection of all control signals generated by C.U. at once.

sequence of signals :-

assume 4 bits

AR	DR	Memory	IR	SP	PC	S	T	GPRs	AC	ALU
In out	In out	Read Write	In out	In out	In out Inc	In out	In out	In out ...	In out	function code

$AR \leftarrow PC$

ND operation

1 0 0 0 0 0 0 0 0 0 0 1 0 0 0 0 0 ... 0 0 0 0 0 0

Control word

$DR \leftarrow M[AR]$

0 1 1 0 1 0 0 0 0 0 0 0 0 0 0 ... 0 0 0 0 0 0



Topic : Control Unit Organization

How control unit generates signals.

Type :—

1. Hardwired control unit
2. Micro-programmed —11—



Topic : Hardwired Control Unit

Control logic is implemented with Gates, flip-flops, decoders and other digital circuits.

Advantage: Can be optimized to produce a faster mode of operation.

Disadvantage: Rearranging the wires among various components is difficult.

→ updation in control logic is difficult

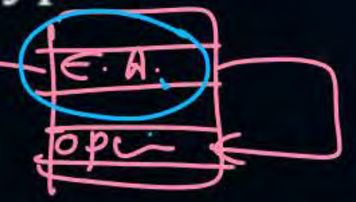
→ suitable only for simple CPUs.

[NAT]

H.W.

#Q. Consider a 6-words instruction, which is of the following type:

Opcode	Mode1	Mode2	Address1	Address2





The first operand (destination) uses register indirect mode and second operand uses indirect mode. Assume each operand is of size 2 words, each address is of 2 words and main memory takes 30ns for 1 byte access. Further assume that the opcode denotes addition operation which copies result of addition of 2 operands. One word size is 4 bytes. Total time required in:

1. Fetch cycle of instruction
2. Execution cycle of instruction
3. Instruction cycle of instruction

1. Fetch :-

$= 24 * 30ns = 720ns$

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2. Execution cycle :-

	operand 1	operand 2
decode	—	—
EA. Calculat^n & operand fetch	$8 * 30 = 240ns$	$(8 * 30ns) + (8 * 30ns) = 480ns$
Execution	—	—
write back result	$8 * 30 = 240ns$	—

Total = 960 ns



2 mins Summary

Topic Datapath

Topic Control Unit Organization

Topic Hardwired Control Unit



Happy Learning

THANK - YOU